

Fetal ultrasound standard plane classification via xception enhancement and brain storm optimization-based feature selection ^{*}

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ARTICLE INFO

Keywords:

Fetal ultrasound
Xception
Feature selection
Brain storm optimization
Classification

ABSTRACT

The reliable classification of fetal ultrasound standard planes is essential for prenatal diagnosis, yet it is complicated by significant inter-class similarity, pervasive speckle noise, and a common scarcity of annotated data. While deep learning offers promising solutions, existing methods often struggle with feature redundancy and loss of discriminative spatial information. To address these limitations, this study introduces a novel hybrid framework that synergistically combines an enhanced deep learning architecture with an advanced optimization-based feature selection strategy. The proposed method first employs an enhanced Xception model, integrating dilated convolutions, combined spatial and channel attention mechanisms, and a Avg-TopK pooling layer to extract rich, multi-scale, and discriminative features. Subsequently, a feature selection algorithm named Brain Storm Optimization with Nearest-Better Clustering (BSOFS-NBC) is applied to refine the feature dimensionality, effectively eliminating redundancy while preserving diagnostically critical information. Evaluated on a large-scale public dataset of 6,630 annotated fetal ultrasound images, our framework demonstrates superior performance, achieving a state-of-the-art classification accuracy of 98.15% with a support vector machine classifier. Comparative analyses confirm consistent and significant improvements in accuracy, precision, recall, and F1-score over established convolutional neural networks (VGG-19, ResNet-101, MobileNet-v2) and recent ensemble methods. These results, supported by high specificity of 99.59%, underscore the effectiveness of our approach and its strong potential to enhance automated prenatal screening systems.

1. Introduction

Irregularities in prenatal development remain a major contributor to perinatal mortality across both developed and developing regions. The World Health Organization (WHO) identifies multiple influencing factors, such as genetic predispositions, maternal health conditions, pregnancy-related complications, and broader socioeconomic and environmental factors, as contributors to restricted fetal growth. Timely detection of abnormal fetal development has the potential to significantly reduce perinatal mortality rates. For effective monitoring of fetal growth, clinicians must accurately identify key fetal biometric measurements, such as abdominal circumference, femur length, and head circumference. These parameters play a critical role in estimating fetal weight and assessing developmental progress. To obtain these measurements, it is necessary to first classify and localize specific fetal organs

and structures, including the fetal brain, abdomen, thorax, and femur (Guo et al., 2023; Salomon et al., 2019). Standardized ultrasound imaging protocols are employed to acquire these views, using established anatomical planes. For a more detailed neurodevelopmental assessment, specialized sub-planes, namely the transventricular, transthalamic, and transcerebellar views, are utilized to capture distinct regions of the fetal brain (Xie et al., 2020).

Currently, deep learning (DL) has emerged as a powerful tool capable of automating the classification of standard planes in ultrasound images. By learning complex patterns directly from data, DL models can provide more consistent, objective, and accurate interpretations compared to manual methods (Fiorentino et al., 2023). Moreover, DL systems can operate at scale, supporting real-time decision-making and improving workflow efficiency in clinical practice. Among DL architectures, convolutional neural networks (CNNs) have been extensively employed

^{*} This work is partly supported by the Natural Science Foundation of Jiangsu Province (Grant No. BK20230419), the Natural Science Foundation of the Jiangsu Higher Education Institutions of China (Grant No. 23KJB520018), and the Zhejiang Provincial Natural Science Foundation of China (Grant No. LQN25F030010).

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<https://doi.org/10.1016/j.eswa.2026.131996>

Received 15 July 2025; Received in revised form 24 February 2026; Accepted 6 March 2026

Available online 12 March 2026

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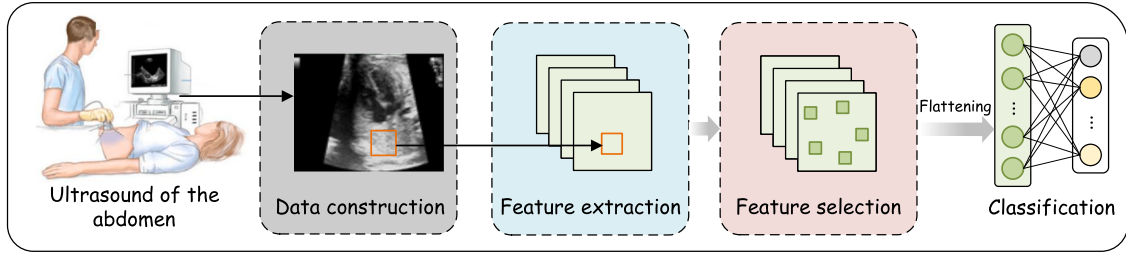


Fig. 1. Pipeline of an automated fetal ultrasound standard plane classification system.

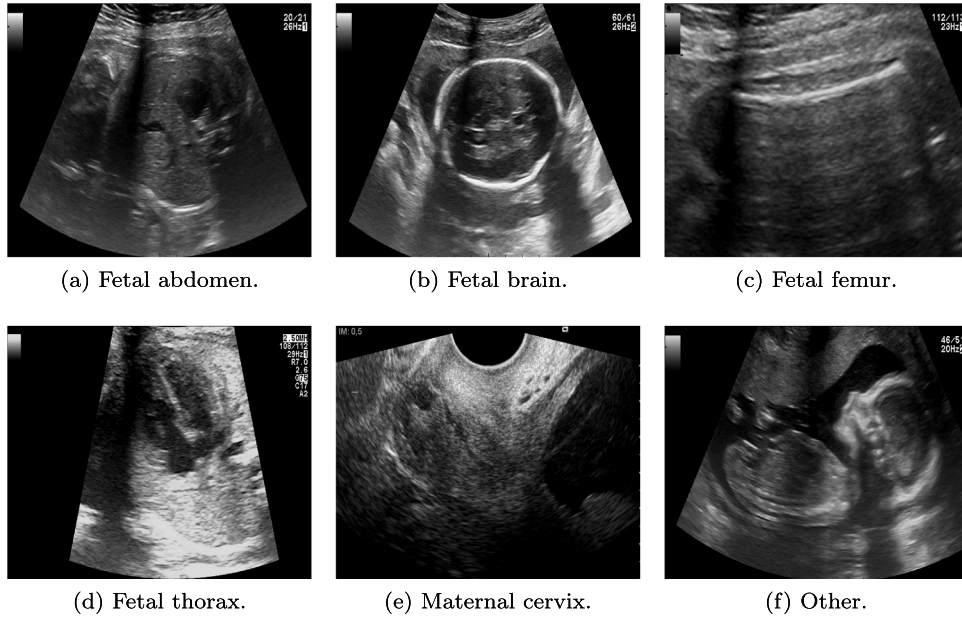


Fig. 2. Representative categories of standard fetal ultrasound scans from the dataset.

in fetal ultrasound analysis due to their ability to capture hierarchical anatomical structures and local discriminative features, leading to high diagnostic accuracy (Pratap et al., 2025; Sridar et al., 2019; Yu et al., 2018). Recent advancements, such as squeeze-and-excitation networks (Qiao & Zulkernine, 2020), convolutional block attention modules (Huo et al., 2024; Turkan et al., 2025), and transformer-based architectures (Sarker et al., 2023), further enhance spatial-semantic representations and improve overall performance. However, CNN-based methods often produce high-dimensional feature representations, increasing the risk of overfitting and hindering interpretability. To address these challenges, hybrid frameworks have been developed that decouple feature extraction, feature selection, and classification into separate stages (Arora & Agarwal, 2024; Gurukala & Verma, 2024; Robindro et al., 2024), as illustrated in Fig. 1. Nevertheless, feature extraction techniques frequently suffer from degradation in spatial fidelity, impairing the model's ability to detect subtle yet clinically significant patterns (Azizi et al., 2021). Moreover, although feature selection is essential for reducing computational complexity in high-dimensional spaces, it may inadvertently discard diagnostically important information (Cömert et al., 2019). These losses compromise the semantic richness and discriminative power of the learned representations. Therefore, there is a critical need for advanced feature extraction paradigms that preserve high-resolution spatial details alongside adaptive dimensionality reduction strategies that retain salient diagnostic features.

Accordingly, the pipeline shown in Fig. 1 for fetal ultrasound standard plane classification faces two key challenges. *Challenge 1*: How can a custom-designed convolutional architecture be optimized to extract semantically rich and spatially discriminative features specific to fetal

standard planes, given the fine-grained inter-class similarities? *Challenge 2*: What feature selection strategies can effectively reduce redundancy while preserving diagnostically relevant features in high-dimensional ultrasound representations, especially when different fetal plane categories exhibit substantial visual overlap?

In this paper, we propose a novel algorithm for fetal ultrasound standard plane classification to address the aforementioned challenges. First, we design an enhanced Xception-based architecture for feature extraction. Then, we develop an improved brain storm optimization (BSO) algorithm for feature selection. Finally, we evaluate the performance of the proposed approach using multiple machine learning classifiers. The main contributions of this work are summarized as follows:

1. To extract semantically rich and spatially discriminative features specific to fetal standard planes, particularly in the presence of fine-grained inter-class similarities, we develop an enhanced Xception-based architecture. This architecture integrates a Avg-TopK pooling strategy, which averages multiple high-activation responses to preserve critical semantic information while reducing noise sensitivity. Additionally, dilated convolutions are incorporated to expand the receptive field without increasing the number of parameters, thereby capturing broader contextual features. A combined spatial and channel attention mechanism is further integrated to emphasize diagnostically relevant regions and channels, enhancing feature discrimination among visually similar planes.
2. To reduce redundancy and retain diagnostically significant features in high-dimensional ultrasound representations, we propose a new feature selection algorithm named BSOFs-NBC. This algorithm en-

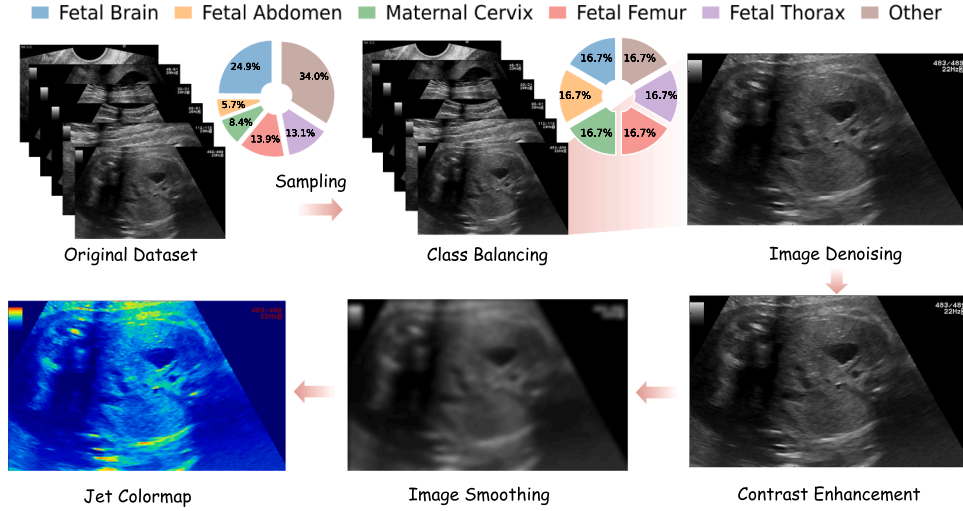


Fig. 3. Data preprocessing and class distribution pipeline. The systematic workflow comprises three sequential stages: (1) Original dataset presentation, displaying the initial imbalanced class distribution across fetal anatomical planes; (2) Strategic sampling, implementing class balancing methodologies to mitigate dataset imbalance; and (3) Comprehensive image preprocessing, employing enhancement protocols including image denoising, image smoothing, contrast enhancement, and jet colormap application to optimize image quality and analytical interpretability for subsequent computational analysis.

codes feature subsets as binary vectors, employs nearest better clustering (NBC) for leader assignment, and adopts leader-guided crossover with adaptive mutation to improve exploration capabilities. BSOFS-NBC effectively identifies discriminative feature subsets, eliminates irrelevant or redundant features, and enhances overall classification performance, particularly in scenarios with substantial visual overlap among plane categories.

Experiments were conducted on the publicly available fetal ultrasound dataset described in [Burgos-Artiz et al. \(2020\)](#). Compared with several state-of-the-art algorithms, including standard Xception, EfficientNet-B0, and ResNet50, the proposed method achieved the highest classification accuracy of 98.15%, demonstrating competitive performance.

The remainder of this article is structured as follows. [Section 2](#) reviews the background and related work on standard plane classification in fetal ultrasound imaging. [Section 3](#) details the proposed method. [Section 4](#) describes the experimental setup and presents the results. Finally, [Section 6](#) concludes the paper and outlines future directions.

2. Related work

In this section, we first introduce fetal ultrasound standard plane classification and review existing approaches CNN-based architectures. We then define the feature selection problem and discuss related work, with a particular focus on population-based stochastic search algorithms.

2.1. Fetal ultrasound standard plane classification

Let $D = \{(x_i, y_i)\}_{i=1}^N$ be a dataset of N fetal ultrasound images, where each $x_i \in \mathbb{R}^{H \times W \times C}$ denotes the i th input image with height H , width W , and C channels. For grayscale images, C equals 1. The corresponding ground truth label is given by $y_i \in \mathcal{Y}$, which includes fetal abdomen, fetal brain, fetal femur, fetal thorax, maternal cervix, and other, as illustrated in [Fig. 2](#). The objective is to classify each image x_i into its correct fetal plane category $y_i \in \mathcal{Y}$ ([Qu et al., 2020](#)).

A deep neural network $\Psi_\theta : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^d$, with learnable parameters θ , maps each input image to a d -dimensional feature vector. A classifier $h : \mathbb{R}^d \rightarrow \mathcal{Y}$ then predicts the final label so that $h(\Psi_\theta(x_i))$ matches

y_i ([Fig. 2](#)). Recent advances in deep learning have enabled automated systems for classifying standard fetal anatomical planes. For example, an early CNN-based approach achieved real-time detection of 13 standard fetal views using weak supervision with image-level labels, reporting an average F1-score of 0.798 and over 90% retrieval accuracy in retrospective evaluations ([Baumgartner et al., 2017](#)). A domain-transferred CNN was proposed for localizing the fetal abdominal standard plane (FASP) in ultrasound videos. By applying transfer learning from CNNs pretrained on natural images, the method effectively reduced overfitting and outperformed traditional low-level feature-based approaches, demonstrating strong potential in data-scarce medical imaging tasks. ([Chen et al., 2015](#)).

In [Zhao et al. \(2024\)](#), a Transfer Learning (TL) approach was introduced to address challenges associated with insufficient training data and reduced model performance. The technique utilizes the prior knowledge embedded in Convolutional Neural Network (CNN) architectures that have been extensively trained on large-scale datasets of natural images. These pretrained models are then adapted to the domain of medical imaging, specifically for the localization of fetal abdominal standard planes. The evaluation of the auxiliary task-plane recognition showed that the integration of Deep CNNs with TL significantly improved performance metrics. In segmentation-related tasks, architectures such as ECAU-Net incorporated efficient channel attention modules to reduce model complexity while achieving high segmentation performance, with Dice and Jaccard scores of 91.35% and 86.01%, respectively ([Shu et al., 2022](#)).

Attention mechanisms have also demonstrated significant effectiveness. The VGGSA model introduced spatial attention into the VGG-19 backbone, enhancing the network's focus on discriminative regions and improving classification performance across multiple datasets ([Krishna & Kokil, 2024a](#)). Multi-task learning frameworks, such as USPD, integrated hybrid knowledge graphs to jointly guide classification and localization, thereby improving both interpretability and accuracy ([Zhao et al., 2022](#)). Furthermore, adaptive attention modules, including adaptive channel weighting combined with multiscale attention mechanisms, have been integrated into pre-trained CNNs to dynamically emphasize diagnostically significant channels, leading to classification accuracies often exceeding 95% ([Krishna & Kokil, 2024b](#)).

Finally, ensemble strategies combining outputs from models such as InceptionResNetV2, DenseNet121, and Xception have been shown to

further enhance robustness and reliability in clinical fetal plane recognition tasks (Krishna & Kokil, 2024c).

2.2. Feature selection

End-to-end deep learning models often generate high-dimensional and redundant features, which can lead to overfitting and reduced interpretability. This issue is particularly pronounced in medical imaging, where annotated data are limited. To address these challenges, modular pipelines that separate feature extraction, feature selection, and classification have gained popularity (Rathika et al., 2024). In such pipelines, CNNs first extract features, feature selection modules reduce redundancy and emphasize diagnostically relevant patterns, and classifiers produce the final predictions. This approach enhances both robustness and interpretability.

Let $\mathcal{X}$ represent the full set of features extracted by the CNN, and let $S \subseteq \mathcal{X}$ be the selected subset such that $|S| \leq k$, where k is a predefined upper limit on the number of features. The feature selection process aims to find an optimal subset S^* that maximizes a performance-oriented objective function $\mathcal{J}(S)$, commonly defined in terms of classification accuracy or F1-score:

$$S^* = \arg \max_{S \subseteq \mathcal{X}, |S| \leq k} \mathcal{J}(S) \quad (1)$$

To further improve predictive performance, a joint optimization framework can be formulated over the network parameters θ and the selected feature subset S . The goal is to maximize $\mathcal{J}(h(f_\theta(x_i)[S]))$, where $\Psi_\theta(x_i)[S]$ denotes the subvector corresponding to the selected features.

Early feature selection methods relied primarily on statistical filter techniques such as correlation coefficients (Liang et al., 2024), mutual information (Li et al., 2024), and the ANOVA F-test (Ragab, 2023). These approaches evaluate individual feature relevance and redundancy efficiently but often fail to capture complex dependencies and interactions among features, leading to suboptimal selections (Zhang et al., 2019). To address these limitations, wrapper-based strategies have been developed (Chaudhuri, 2024). These methods evaluate subsets directly by measuring classification performance with classifiers such as random forest, support vector machines, and artificial neural networks. While wrapper methods generally achieve better results, they are computationally expensive, particularly as feature dimensionality increases.

In recent years, population-based metaheuristic algorithms, including genetic algorithms (GA) and particle swarm optimization (PSO), have been widely used for feature selection. These algorithms simulate evolutionary and swarm intelligence processes to globally explore the search space, effectively balancing exploration and exploitation to identify near-optimal subsets. Such approaches are especially beneficial for high-dimensional data, improving classification performance while reducing redundancy. For example, PSO combined with machine learning classifiers has been used to reduce feature dimensionality from 30 to 19 while achieving 100% accuracy on the UCI WDBC breast cancer dataset (Kazerani, 2024). In distributed environments, hybrid approaches that integrate PSO with mutual information have achieved over 98% accuracy on imbalanced datasets, including kidney disease classification tasks (Robindro et al., 2024). Brain storm optimization (BSO) has also emerged as a promising alternative. Enhanced variants, such as binary BSO-based feature selection, have demonstrated superior performance compared to traditional methods by incorporating advanced clustering strategies and refined update mechanisms (Zhang et al., 2019). In Pourpanah et al. (2019), a hybrid method combining a fuzzy min-max neural network with BSO was proposed to improve both feature selection and classification. This method incrementally constructs hyperboxes to model data distributions while BSO searches for optimal feature subsets. Experiments on benchmark and real-world datasets showed that this approach outperforms conventional fuzzy min-max networks and other swarm-based algorithms such as PSO and GA.

3. Proposed method

In this section, we present the proposed fetal ultrasound plane classification framework in detail. First, we describe the data preprocessing procedure. We then introduce an enhanced Xception architecture, which integrates dilated convolutions to enlarge the receptive field, as well as channel and spatial attention mechanisms to emphasize diagnostically significant regions and feature channels. Additionally, we incorporate the Avg-TopK pooling strategy to further improve feature robustness and discrimination. Subsequently, we describe an improved brain storm optimization algorithm designed for feature selection. This method employs nearest-better clustering (NBC) as the primary clustering strategy and adopts a leader-guided mutation mechanism to enhance exploration and avoid local optima. Finally, we provide a comprehensive overview of the full training process for fetal standard plane classification using ultrasound imaging.

3.1. Data preprocessing

We employed a publicly available fetal ultrasound plane image dataset comprising 12,400 images (Burgos-Artiz et al., 2020). Before algorithm development, data preprocessing was performed to ensure quality and consistency. This included class balancing to address potential data imbalance, image denoising to remove artifacts, contrast enhancement to improve structural differentiation, and image smoothing to reduce noise while preserving critical anatomical details. Additionally, a Jet colormap was applied to selected samples for visualization purposes, facilitating qualitative assessment and highlighting enhanced structural contrasts. The overall data preprocessing workflow is illustrated in Fig. 3.

Class balancing. The dataset comprises 711 images of the fetal abdomen, 3,092 of the fetal brain, 1,626 of the maternal cervix, 1,040 of the fetal femur, 1,718 of the fetal thorax, and 4,213 categorized as others. The dataset was first partitioned into training, validation and testing subsets. To address class imbalance, class balancing techniques were applied exclusively to the training subset after data separation. A target size of 1,105 samples per class was defined based on the minority class distribution and experimental considerations (Bria et al., 2020). Within the training data, the synthetic minority oversampling technique (SMOTE) was applied to underrepresented classes (fetal abdomen and fetal femur), generating synthetic samples through interpolation between existing minority instances. For overrepresented classes (fetal brain, maternal cervix, fetal thorax, and others), random under-sampling was performed to match the target distribution. The validation set and testing set were kept completely untouched and contained only original images to prevent data leakage and ensure unbiased evaluation. All undersampling operations were conducted using a fixed random seed to ensure reproducibility.

Image denoising. Fetal ultrasound images are frequently degraded by speckle noise, an artifact caused by signal interference in coherent imaging systems. This noise obscures fine anatomical details and hinders accurate identification of standard fetal planes, which is crucial for prenatal diagnosis. Conventional denoising methods such as Gaussian or median filtering often fail to balance noise suppression with edge preservation, resulting in blurred features or loss of important texture needed for classification. To overcome these limitations, we employ wavelet denoising, a multi-resolution technique that separates noise from signal components across different frequency bands. Wavelet transforms analyze image features at multiple scales, enabling precise localization of noise and preservation of edges and fine structures like the fetal brain or femur. The improved image clarity enhances the reliability of subsequent feature extraction and classification, as supported by prior studies (Yan et al., 2021). By reducing noise without compromising anatomical detail, wavelet denoising directly improves the accuracy of automated

fetal plane classification and reduces diagnostic uncertainty in clinical practice.

Contrast enhancement. Fetal ultrasound images often exhibit low contrast and uneven illumination, which obscure subtle anatomical details and complicate accurate interpretation. Traditional methods like histogram equalization tend to over-amplify noise or distort intensity distributions, reducing diagnostic reliability. To overcome these issues, adaptive Gamma correction with weighted distribution (AGCWD) adjusts brightness dynamically using localized histogram statistics, preserving natural intensity transitions and selectively enhancing poorly illuminated regions. This approach improves feature visibility without introducing artifacts, as demonstrated in Rahman et al. (2016), thereby enabling more robust feature extraction and classification and enhancing the accuracy of automated fetal plane diagnosis.

Image smoothing. Fetal ultrasound images frequently contain high-frequency noise and artifacts that obscure anatomical details and degrade overall image quality, affecting diagnostic accuracy. Traditional smoothing techniques often fail to adequately suppress noise without blurring important structures. Gaussian blur addresses this by applying a weighted filter based on the Gaussian distribution, which selectively reduces noise while maintaining key structural boundaries (Honarvar Shakibaei Asli et al., 2021). This results in cleaner and more uniform images, enhancing the reliability of subsequent segmentation, feature extraction, and classification, and ultimately improving diagnostic outcomes.

3.2. Enhanced Xception architecture for feature extraction

In this study, we propose an enhanced Xception architecture specifically designed for feature extraction in fetal ultrasound standard plane classification. The overall network architecture is illustrated in Fig. 4, with the improved components highlighted in red. These modifications aim to better capture subtle anatomical details and improve discriminative performance in noisy and complex ultrasound environments.

First, we replace the conventional convolutional layer in the entry flow with a 3×3 dilated convolution. This modification expands the receptive field without increasing parameter count or computational cost, enabling the network to capture broader contextual information critical for identifying anatomical structures. The dilated convolution facilitates multi-scale feature extraction, which is particularly advantageous for handling variable fetal poses and small yet diagnostically important anatomical cues that are often obscured by speckle noise. Second, we incorporate both channel attention and spatial attention mechanisms into the entry flow. The channel attention module, configured with 256 intermediate features, adaptively recalibrates channel-wise feature importance, enhancing the representation of meaningful features. The spatial attention module further refines the focus by emphasizing informative regions while suppressing irrelevant background artifacts. Together, these attention mechanisms enable the network to selectively concentrate on diagnostically significant areas, thereby improving feature discrimination in standard plane classification. Third, we introduce an Avg-TopK pooling layer with a 3×3 kernel to replace traditional max pooling. This strategy retains the top-K activations within each region, effectively preserving high-importance responses and mitigating the loss of critical spatial information that often occurs with aggressive pooling. By prioritizing the most salient features, Avg-TopK pooling enhances global feature representation and contributes to improved classification accuracy.

The following subsections provide a detailed description of each modification, including architectural configurations and the design motivations.

3.2.1. Dilated convolution

In the original Xception architecture, the entry block begins with separable convolutional layers designed to achieve high computational

efficiency while maintaining strong representational power. However, these layers have a limited receptive field, which constrains the model's ability to capture long-range spatial dependencies critical for accurate fetal ultrasound standard plane classification. To address this limitation, we replace the first separable convolution with a dilated convolutional layer.

Dilated convolution, also known as atrous convolution, expands the receptive field by inserting gaps between kernel elements. This design enables the model to aggregate multi-scale contextual information without increasing the number of parameters or computational cost (Wei et al., 2018). Formally, a one-dimensional dilated convolution on a signal x with a kernel w of size K and a dilation rate r is defined as

$$y[i] = \sum_{j=0}^{K-1} x[i + r \cdot j] \cdot w[j], \quad (2)$$

where K denotes the kernel size, r is the dilation rate, and j is the summation index. This formulation extends directly to two-dimensional convolutions, effectively expanding the kernel's coverage without increasing parameter count. The effective receptive field can be quantified as $K + (K - 1)(r - 1)$, allowing the kernel to span a larger area while preserving resolution.

In fetal ultrasound images, anatomical structures such as the brain, thorax, and abdomen often exhibit low contrast, indistinct boundaries, and significant speckle noise. By expanding the receptive field early in the network, the dilated convolution facilitates the integration of broader anatomical context, leading to enhanced feature representation and improved discrimination of standard planes. This modification improves robustness and generalization in challenging clinical scenarios where subtle spatial relationships are critical for accurate classification.

3.2.2. Attention mechanisms

The original Xception architecture applies separable convolutions uniformly across all channels and spatial locations, limiting the network's ability to prioritize diagnostically relevant features in fetal ultrasound images. To enhance feature discrimination and spatial focus, we replace the fourth and sixth separable convolutional layers in the entry block with channel attention and spatial attention modules, respectively.

Channel attention adaptively recalibrates each feature channel's importance, enhancing representation of discriminative channels corresponding to subtle anatomical details such as the fetal brain or abdomen. Given input feature map $\mathbf{F} \in \mathbb{R}^{H \times W \times C}$, where H , W , and C denote height, width, and number of channels, channel attention is computed by applying global average pooling (GAP) and global max pooling (GMP) along spatial dimensions, yielding descriptors $\mathbf{F}_{\text{avg}}$ and $\mathbf{F}_{\text{max}} \in \mathbb{R}^{1 \times 1 \times C}$. These pass through a shared multilayer perceptron (MLP) with ReLU activation and reduction ratio r , producing $\mathbf{M}_{\text{avg}}$ and $\mathbf{M}_{\text{max}}$. The combined output is activated by a sigmoid function to form the channel attention map $\mathbf{M}_c$. The refined feature map $\mathbf{F}'$ results from element-wise multiplication of $\mathbf{M}_c$ and $\mathbf{F}$.

$$\mathbf{F}_{\text{avg}} = \text{GAP}(\mathbf{F}), \quad \mathbf{F}_{\text{max}} = \text{GMP}(\mathbf{F}) \quad (3)$$

$$\mathbf{M}_{\text{avg}} = \mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{F}_{\text{avg}}), \quad \mathbf{M}_{\text{max}} = \mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{F}_{\text{max}}) \quad (4)$$

$$\mathbf{M}_c = \sigma(\mathbf{M}_{\text{avg}} + \mathbf{M}_{\text{max}}) \quad (5)$$

$$\mathbf{F}' = \mathbf{M}_c \otimes \mathbf{F} \quad (6)$$

where $\mathbf{W}_1 \in \mathbb{R}^{\frac{C}{r} \times C}$ and $\mathbf{W}_2 \in \mathbb{R}^{C \times \frac{C}{r}}$ are the weight matrices of the MLP, $\sigma(\cdot)$ denotes the sigmoid function, and $\otimes$ indicates element-wise multiplication broadcast spatially.

Spatial attention highlights informative spatial regions crucial for localizing fetal anatomical structures with variable shape and position. Given $\mathbf{F}' \in \mathbb{R}^{H \times W \times C}$, spatial attention aggregates channel-wise information via channel-wise average pooling (CAP) and max pooling (CMP), producing $\mathbf{F}_{\text{avg}}^s$ and $\mathbf{F}_{\text{max}}^s \in \mathbb{R}^{H \times W \times 1}$. These are concatenated and convolved with a 7×7 kernel, followed by a sigmoid activation, to yield

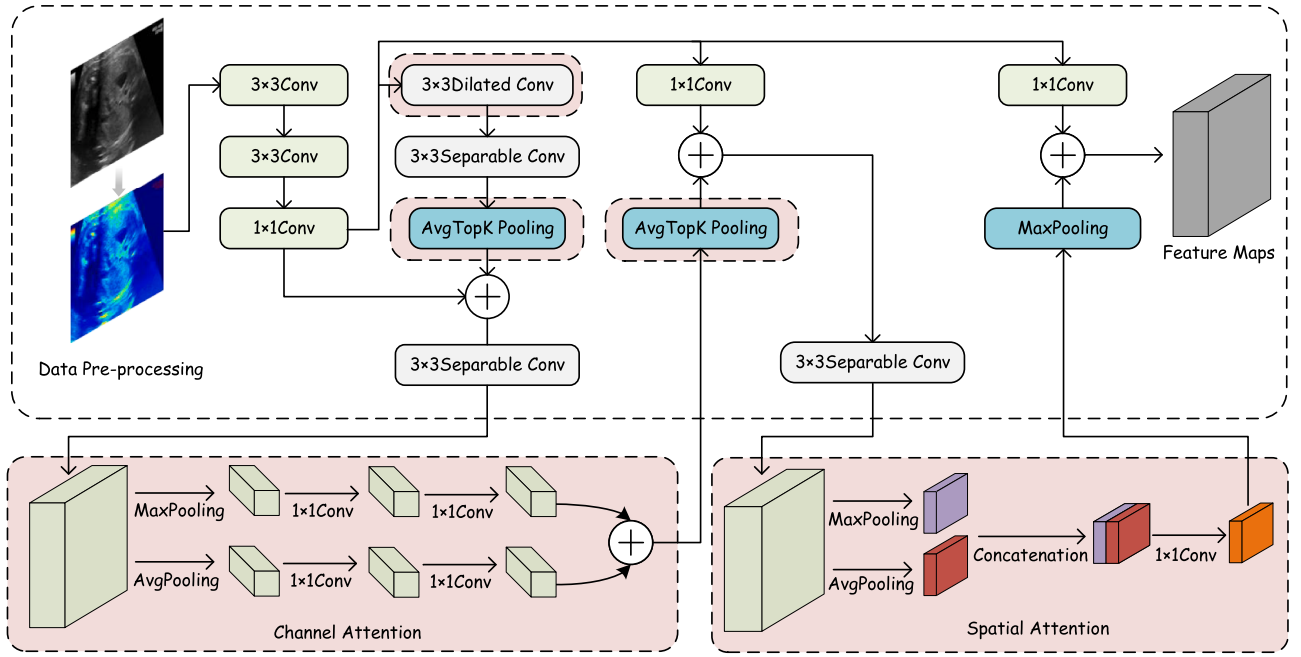


Fig. 4. The architecture of the proposed enhanced Xception model for feature extraction. Key modifications from the original Xception are highlighted in red and include: (a) a dilated convolution in the entry flow to expand the receptive field; (b) integrated channel and spatial attention modules to emphasize diagnostically significant features and regions; and (c) an Avg-TopK pooling layer to preserve high-activation responses while mitigating noise. A residual connection is maintained to facilitate gradient flow during training.

the spatial attention map M_s . The output F'' is computed by element-wise multiplication of M_s and F' .

$$F_{avg}^s = CAP(F'), \quad F_{max}^s = CMP(F') \quad (7)$$

$$F^s = \text{Concat}[F_{avg}^s, F_{max}^s] \quad (8)$$

$$M_s = \sigma(\text{Conv}_{7 \times 7}(F^s)) \quad (9)$$

$$F'' = M_s \otimes F' \quad (10)$$

where $\text{Conv}_{7 \times 7}$ is a convolution with a 7×7 kernel and $\sigma(\cdot)$ is the sigmoid activation. Element-wise multiplication $\otimes$ is broadcast along the channel dimension.

By integrating channel and spatial attention, the network learns both what and where to emphasize, improving discrimination and spatial localization in noisy, low-contrast fetal ultrasound images. This enhances robustness and classification accuracy, particularly for visually similar standard planes.

3.2.3. Avg-TopK pooling

In the original Xception entry block, each separable convolutional pair is followed by max pooling to reduce spatial dimensions and retain dominant features. In our design, we replace the first two max pooling layers with Avg-TopK pooling, which computes the average of the top-K activations within a local region (Özdemir, 2023). Traditional pooling methods have inherent limitations. Max pooling preserves only the single highest activation, potentially discarding other informative responses. Average pooling treats all activations equally, which can dilute dominant features when low-magnitude or noisy values are present, biasing the average toward zero (Walter, 2023; You et al., 2021). Both methods are inadequate for emphasizing truly discriminative features in noisy fetal ultrasound data.

To overcome these drawbacks, Avg-TopK pooling averages the top-K highest activations rather than relying solely on the maximum or the mean of all values. This approach retains significant features by considering multiple high-magnitude activations while reducing the influence of noise and near-zero values. It also avoids the noise amplification risk associated with max pooling when the highest activation corresponds to noise (Hossain et al., 2023).

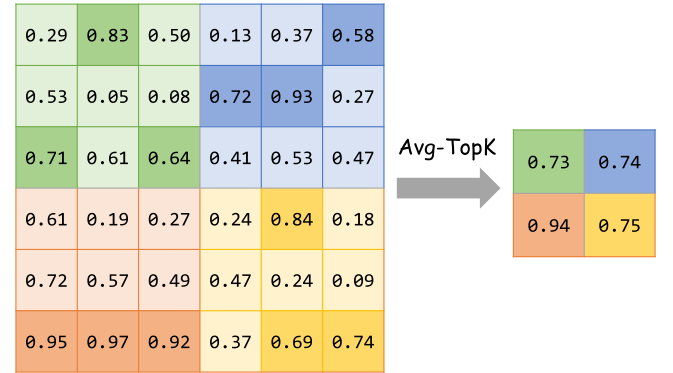


Fig. 5. Example of Avg-TopK pooling, where a 6×6 feature map as input, using a 3×3 pooling kernel with a stride of 3; the highlighted regions indicate the top-3 values within each window.

Formally, let $X = \{x_1, x_2, \dots, x_n\}$ denote the set of pixel values within the pooling region from a convolutional layer's output. Sorting these values in descending order yields.

$$\text{sort}(X) = \{x_{(1)}, x_{(2)}, \dots, x_{(n)}\}, \quad \text{where } x_{(1)} \geq x_{(2)} \geq \dots \geq x_{(n)}.$$

The Avg-TopK pooling output is then defined as the average of the top K values:

$$X_{\text{AvgTopK}} = \frac{1}{K} \sum_{i=1}^K x_{(i)}.$$

This approach provides a robust and balanced feature representation by combining the strengths of max and average pooling while mitigating their respective weaknesses. Fig. 5 illustrates the Avg-TopK pooling process with $K = 3$ applied to a 6×6 input using a 3×3 pooling kernel.

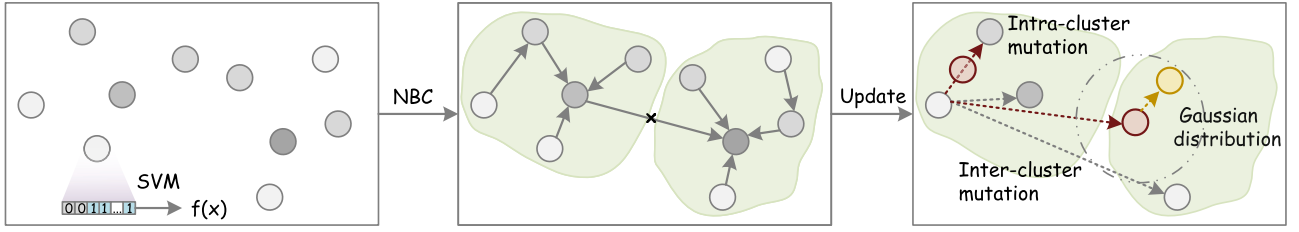


Fig. 6. Brain storm optimization-based feature selection using nearest better clustering workflow. The schematic illustrates the complete optimization process comprising three key stages: (1) Random binary feature subset generation; (2) Grouping solutions based on nearest better clustering; (3) Intra-cluster recombination, inter-cluster recombination, and Gaussian perturbation operations.

3.3. Brain storm optimization with nearest-better clustering for feature selection

To perform feature selection, we propose a novel variant of the brain storm optimization (BSO) algorithm, named BSOFS-NBC. Inspired by human brainstorming processes, BSO clusters candidate solutions and iteratively updates the population through inter-cluster and intra-cluster mutations. This clustering mechanism enhances population diversity and significantly improves global search capability compared to conventional population-based random search algorithms.

The original BSO algorithm (Shi, 2011) was designed for continuous optimization problems. However, a major limitation of this approach is that it does not utilize fitness information, making it ineffective at identifying closely related individuals within the fitness landscape. As a result, the original BSO is not directly applicable to combinatorial tasks such as feature selection. To address this limitation, BSOFS-NBC introduces a binary encoding scheme to represent feature subsets and integrates nearest-better clustering (NBC) (Luo et al., 2021), which adaptively captures the fitness landscape and facilitates effective escape from local optima. To guide the search toward highly discriminative subsets, a support vector machine (SVM) classifier is used to evaluate candidate feature sets, providing reliable feedback on their classification performance.

Fig. 6 illustrates the overall framework of BSOFS-NBC. The detailed procedure is outlined in Algorithm 1, with each step described as follows.

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First, a binary population $P = \{x_1, x_2, \dots, x_N\}$, $x_i \in \{0, 1\}^D$ with size N is initialized, where each individual is represented as a binary vector x of dimension D , indicating whether each feature is selected (1) or not (0) (Line 1). The fitness of each individual is evaluated using SVM classifier with a radial basis function (RBF) kernel (Line 2). Each binary vector defines a subset of features used to train the SVM on the training data, and the accuracy on a validation set is taken as the fitness value. Formally,

$$\mathcal{J}(x) = \begin{cases} \mathcal{A}(x), & \text{if } \|x\|_0 > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

where $\mathcal{A}(x)$ denotes the classification accuracy on the validation set obtained by training a classifier using the feature subset specified by x .

In each iteration, NBC is applied to cluster the population into distinct species and assign leaders (Line 4). The detailed procedure of NBC is summarized in Algorithm 2. Given that individuals are encoded as binary vectors, the distance between individuals in the solution space is measured using the Hamming distance. For each individual x_i , all individuals with higher fitness values $P_{i,better}$ are first identified. Among these, the individual with the smallest Hamming distance to x_i is selected as its nearest better neighbor, denoted as its leader $x_{i,leader}$.

After establishing these nearest-better relationships, a directed graph is constructed in which the best individual serves as the root node, and each remaining individual has a single edge pointing to its leader. By removing the $k - 1$ longest edges in this graph, the population is parti-

Algorithm 1 BSOFS-NBC.

Require: Population size N , feature dimension D , max iterations T_{max}
Ensure: Best feature subset $x^* \in \{0, 1\}^D$

- 1: Initialize population $P = \{x_1, x_2, \dots, x_N\}$, $x_i \in \{0, 1\}^D$
- 2: Evaluate fitness $f(x_i)$ for all x_i ▷ Eq. (11)
- 3: Initialize best solution $x^* \leftarrow \arg \max_{x_i \in P} f(x_i)$
- 4: **for** $t \leftarrow 1$ to T_{max} **do**
- 5: clusters $\leftarrow \text{NBC}(P, f, D)$ ▷ Algorithm 2
- 6: **for** $i \leftarrow 1$ to N **do**
- 7: Select random $x_{r_1} \in P$
- 8: **if** $U(0, 1) < p_c$ **then**
- 9: $x'_i \leftarrow (1 - r) \cdot x_{r_1} + r \cdot x_{r_1,leader}$ ▷ Intra-cluster recombination
- 10: **else**
- 11: Select random $x_j \in P$, $x_{k_1} \in C_{k_1}$, $x_{k_2} \in C_{k_2}$
- 12: $x'_i \leftarrow (1 - r_1 - r_2) \cdot x_j + r_1 \cdot x_{k_1} + r_2 \cdot x_{k_2}$ ▷ Inter-cluster recombination
- 13: **end if**
- 14: Apply Gaussian perturbation to x'_i ▷ Eq. (13)
- 15: Binarize x'_i to generate final offspring ▷ Eq. (14)
- 16: **end for**
- 17: Evaluate fitness for all offspring
- 18: Replace x_i with x'_i if $f(x'_i) > f(x_i)$
- 19: Update best solution x^* if improved
- 20: **end for**
- 21: **Return** x^*

tioned into k trees, forming a forest. Each tree represents a species, and its root node is designated as the center of that species.

Offspring solutions are generated through recombination between a leader and one or more randomly selected individuals. The new candidate x'_i is constructed as

$$x'_i = \begin{cases} (1 - r) \cdot x_{r_1} + r \cdot x_{r_1,leader}, & \text{if } U(0, 1) < p_c, \\ (1 - r_1 - r_2) \cdot x_j + r_1 \cdot x_{k_1} + r_2 \cdot x_{k_2}, & \text{otherwise,} \end{cases} \quad (12)$$

where $r, r_1, r_2 \in [0, 1]^D$ are random weight vectors, x_{r_1} is a randomly chosen individual from the same cluster, $x_{r_1,leader}$ denotes its leader, x_j is a base individual randomly selected from the population, and x_{k_1}, x_{k_2} are selected from two clusters, respectively. The parameter p_c is a uniformly sampled random value in $[0, 1]$, and p_c denotes the probability threshold that determines whether intra-cluster or inter-cluster recombination is applied.

After recombination, mutation is applied using Gaussian perturbation scaled by a logistic-decayed factor (Line 14). The step size is defined as

$$s_t = \sigma\left(\frac{0.5 \cdot T_{max} - t}{20}\right) \cdot \mathcal{N}(0, 1),$$

where $\sigma(z) = \frac{1}{1 + e^{-z}}$ is the logistic sigmoid function, t is the current iteration, and $\mathcal{N}(0, 1)$ denotes Gaussian noise. The updated vector is given

Algorithm 2 Nearest better clustering (NBC).

Require: Population $P = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, fitness values f , Hamming distance matrix D

Ensure: Clusters C

```

1: for  $i \leftarrow 1$  to  $N$  do
2:   Identify set  $P_{i,better} \leftarrow \{\mathbf{x}_j \mid f(\mathbf{x}_j) > f(\mathbf{x}_i)\}$ 
3:   if  $P_{i,better} \neq \emptyset$  then
4:      $\mathbf{x}_{i,leader} \leftarrow \arg \min_{\mathbf{x}_j \in P_{i,better}} \text{Hamming}(\mathbf{x}_i, \mathbf{x}_j)$ 
5:     Store edge from  $\mathbf{x}_i$  to  $\mathbf{x}_{i,leader}$ 
6:   else
7:      $\mathbf{x}_i$  is marked as a root
8:   end if
9: end for
10: Sort all edges by Hamming distance in descending order
11: Remove the  $k - 1$  longest edges to form  $k$  clusters  $C$ 
12: Identify roots as individuals with no incoming edges after edge removal
13: return cluster assignments  $C$ , leader indices  $L$ 

```

by

$$\mathbf{x}'_i = \mathbf{x}'_i + s_i \cdot \mathcal{N}(0, 1). \quad (13)$$

The final binary solution is obtained via a threshold function:

$$\mathbf{x}'_i[j] = \begin{cases} 1, & \text{if } \mathbf{x}'_i[j] > 0.5, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

Each offspring is subsequently evaluated (Line 17). If an offspring demonstrates superior fitness compared to its parent, it replaces the parent in the population (Line 18). The global best solution is updated whenever an improved individual is found (Line 19). This iterative process continues until a predefined stopping criterion is satisfied, such as reaching the maximum number of fitness evaluations or iterations. Finally, the individual achieving the highest validation accuracy is selected as the final feature subset.

3.4. Training procedure

In this section, we present the training pipeline of the proposed model, which follows a three-stage process integrating feature extraction, feature selection, and final classification. First, raw ultrasound images are preprocessed through normalization and data augmentation techniques, including random flipping, rotation, and zooming. The preprocessed images are then passed through the enhanced Xception architecture, which leverages dilated convolutions, attention mechanisms, and a custom Avg-TopK pooling layer to extract semantically rich and contextually discriminative features. Next, the high-dimensional feature representations are refined via feature selection using BSOFS-NBC, which iteratively searches for optimal feature subsets by modeling population-based knowledge transfer among clustered individuals, effectively balancing exploration and exploitation during recombination. The selected features are subsequently standardized using z-score normalization and classified with SVM. The hyperparameters C and γ of the SVM are tuned using grid search with 5-fold cross-validation to maximize validation performance. Finally, the trained model is evaluated on an independent test set to assess its generalization capability.

4. Experiment

In this section, we present a comprehensive evaluation of the proposed method. First, we describe the experimental setup, including the dataset, evaluation metrics, and experimental design. Then, we present a detailed analysis of the proposed model's performance using visualization tools such as confusion matrices and ROC curves, along with a class-wise evaluation across six standard fetal planes. Next, we compare

the proposed method against recent existing approaches to demonstrate its superiority and practical relevance. Finally, we compare the performance of the proposed model with pre-trained CNN architectures and provides an ablation study to highlight the contributions of each component.

4.1. Experimental setup

Dataset. To validate the proposed method, experiments were conducted on the largest publicly available dataset of standard fetal ultrasound planes (Burgos-Artiz et al., 2020). All images were manually labeled by experienced obstetricians and categorized into six classes: fetal abdomen, fetal brain, maternal cervix, fetal femur, fetal thorax, and a miscellaneous "other" class. The dataset consists of 12,400 ultrasound images, including 711 fetal abdomen, 3,092 fetal brain, 1,626 maternal cervix, 1,040 fetal femur, 1,718 fetal thorax, and 4,213 "other" images. The dataset was partitioned into three mutually exclusive subsets, allocating 70% for training, 20% for validation, and 10% for testing purposes. To address the significant class imbalance, class balancing techniques were applied exclusively to the training subset after data separation. Specifically, a target size of 1,105 samples per class was defined based on the minority class distribution and experimental considerations. Oversampling and undersampling strategies were then applied only within the training data to achieve a balanced distribution. The validation set remained completely untouched and contained only original images to prevent data leakage and ensure unbiased evaluation.

Evaluation metrics. Evaluation metrics are critical for assessing the performance of classification models, particularly in medical applications where prediction errors may have serious consequences. Since overall accuracy alone can be misleading in the presence of class imbalance, multiple complementary metrics are employed to provide a comprehensive and reproducible evaluation. In this study, accuracy, precision, recall, F1-score, and specificity are formally defined and computed in a class-wise manner, followed by macro-averaging across all classes to ensure equal contribution from each category.

Let TP_c , TN_c , FP_c , and FN_c denote the number of true positives, true negatives, false positives, and false negatives for class c , respectively. For a C -class classification problem, the evaluation metrics are defined as follows.

Accuracy measures the proportion of correctly classified samples over the entire dataset:

$$\text{Accuracy} = \frac{\sum_{c=1}^C (TP_c + TN_c)}{\sum_{c=1}^C (TP_c + TN_c + FP_c + FN_c)}. \quad (15)$$

Precision quantifies the proportion of correctly predicted samples for each class among all samples assigned to that class:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}. \quad (16)$$

Recall (Sensitivity) measures the proportion of actual samples of class c that are correctly identified:

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}. \quad (17)$$

F1-score is defined as the harmonic mean of Precision and Recall for each class:

$$\text{F1-score}_c = 2 \times \frac{\text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \quad (18)$$

Specificity (True Negative Rate) evaluates the ability of the model to correctly identify negative samples for each class:

$$\text{Specificity}_c = \frac{TN_c}{TN_c + FP_c}. \quad (19)$$

For precision, recall, F1-score, and specificity, macro-averaged results are reported by computing the arithmetic mean across all classes, thereby preventing dominance by majority classes and ensuring fair evaluation under class imbalance.

Table 1
Hyperparameters for model training (left) and feature selection (right).

Model Training		BSOFS-NBC	
Parameter	Value	Parameter	Value
Number of Epochs	30	Population Size (N)	4D
Iterations per Epoch	2,321	Max Function Evaluations (maxFEs)	1000D
Optimizer	SGD	Max Iterations ($T_{\max}$)	maxFEs/ N
Learning Rate	0.001	Probability Threshold (p_c)	0.1
Batch Size	32	Species Size (S_r)	20

Experimental design. All experiments involving the deep learning models and machine learning classifiers were conducted in Python, using essential libraries including TensorFlow, Keras, and Scikit-learn. The experimental procedures were executed on a workstation equipped with an Intel Core i5 processor operating at 2.90 GHz, 64 GB of RAM, and an NVIDIA Quadro RTX 4000 GPU, running Windows 11 x64. The training of CNN architectures was performed using standardized hyperparameter settings to ensure fair comparisons and reproducibility. Table 1 summarizes the key hyperparameters employed during model training and feature selection within BSOFS-NBC.

The CNN models were trained using 70% of the dataset, with the remaining 30% reserved for validation to assess generalization performance. All training processes were repeated multiple times to ensure stability, and the final reported results represent the average performance across runs.

4.2. Experimental results

For the classification, SVM was employed due to its strong capability in handling high-dimensional data and its effectiveness in both binary and multi-class classification scenarios. Fig. 7 presents the confusion matrix of the proposed model, highlighting its classification performance across six standard fetal plane classes, i.e., Fetal Abdomen, Fetal Brain, Fetal Femur, Fetal Thorax, Maternal Cervix, and Other. The model achieves exceptionally high accuracy, with most classes classified perfectly or nearly perfectly, as reflected by the strong diagonal dominance in the matrix. For instance, the Fetal Abdomen and Fetal Brain classes achieved almost perfect true positive counts (330 out of 331), while minimal confusion was observed in the Other category, which occasionally overlapped with classes such as Femur and Thorax.

Fig. 7 shows the receiver operating characteristic (ROC) curves generated using a one-vs-rest strategy for each class. The area under the curve (AUC) scores confirm the excellent discriminative power of the model, reaching 1.00 for five of the six classes and 0.99 for the Other class. These results indicate near-perfect sensitivity and robust performance across all classes.

Table 2 summarizes the class-wise evaluation in terms of precision, recall, F1-score, and specificity. The results demonstrate consistently high scores for all standard fetal plane classes. The Fetal Brain and Maternal Cervix classes achieved perfect or near-perfect values across all metrics. Similarly, the Fetal Femur and Fetal Thorax classes exhibited strong performance, with F1-scores of 96.74% and 97.76%, respectively. Although the Other class showed slightly lower precision and recall due to greater variability, it still maintained a high F1-score of 93.73% and specificity of 99.46%, underscoring the model's robustness in handling diverse and challenging ultrasound data.

Figs. 8 and 9 illustrate the interpretability analysis of the proposed fetal ultrasound classification framework using Grad-CAM and occlusion sensitivity techniques, respectively. These visualizations are essential for building clinical trust, as they provide a qualitative "window into the model's reasoning" and verify that predictions are grounded in medically meaningful features.

The Grad-CAM heatmaps in Fig. 8 generate class-discriminative visual explanations by leveraging the gradients flowing into the final convolutional layer of the enhanced Xception architecture. The resulting

Table 2
Class-wise performance comparison.

Class	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)
Fetal Abdomen	99.55	99.39	99.69	99.54	99.88
Fetal Brain	99.64	100	99.69	99.85	100
Fetal Femur	98.81	96.04	98.49	96.74	98.97
Fetal Thorax	99.27	96.75	98.79	97.76	99.34
Maternal Cervix	99.94	99.70	100	99.85	99.94
Other	96.97	97.03	90.82	93.73	99.46

heatmaps consistently highlight spatial regions that most strongly influence the model's prediction. Crucially, our analysis reveals that these high-activation regions align precisely with known anatomical landmarks that sonographers use for standard plane identification. For instance, in the Fetal Brain class, the model's activation is concentrated around the ventricles and the overall cranial circumference. For the Fetal Femur, the heatmap sharply outlines the long bone structure. In the Fetal Thorax class, the model focuses on the cardiac silhouette and the ribcage. This consistent anatomical alignment demonstrates that the feature extraction and selection pipeline successfully guides the model to learn human-interpretable and clinically relevant features, rather than relying on irrelevant background artifacts or speckle noise.

The occlusion sensitivity maps in Fig. 9 provide a complementary and more direct validation of feature importance. By systematically occluding different regions of the input image with a gray patch and monitoring the subsequent drop in the model's classification confidence, we can identify the image regions that are most critical for the correct prediction. The results strongly corroborate the findings from Grad-CAM. When critical anatomical structures are occluded for example, the femur bone or the brain anatomy a pronounced decrease in prediction confidence is observed, as indicated by the dark regions in the sensitivity maps. Conversely, occluding non-informative areas of the image, such as the background, results in minimal change to the output confidence. This two-pronged interpretability approach, using both Grad-CAM and occlusion sensitivity, offers robust qualitative evidence of the framework's reliability. It confirms that our model not only achieves high numerical accuracy but also makes its decisions based on a semantically meaningful understanding of fetal ultrasound content, thereby enhancing its potential utility and acceptability in a clinical decision-support setting.

4.3. Comparative experimental analysis

Table 3 provides a comprehensive comparison between the proposed method and several state-of-the-art approaches, evaluated on key metrics including accuracy, precision, recall, F1-score, and specificity. The baseline methods include Optimized Feature Selection, Ensemble of Deep Learning, Adaptive Channel Weight, Spatial Attention with VGG-19, Decision Fusion-Based CNN, SonoNet-32, and SonoNet-16, all recognized for their competitive performance in fetal ultrasound classification.

The proposed method consistently outperforms all compared techniques across every metric. It achieves the highest accuracy of 98.15%, demonstrating superior overall classification effectiveness. Precision reaches 98.12%, indicating strong control over false positives, while recall attains 97.72%, reflecting robust true positive identification with minimal false negatives. The F1-score balances these aspects, achieving 98.27%, the best among all methods. Notably, the method attains a specificity of 99.59%, underscoring its exceptional ability to correctly identify negative cases, which is critical in medical diagnostics to reduce false alarms.

These results confirm the proposed method's advantage over individual deep learning models as well as ensemble and attention-based approaches. The consistent improvements across all evaluation criteria

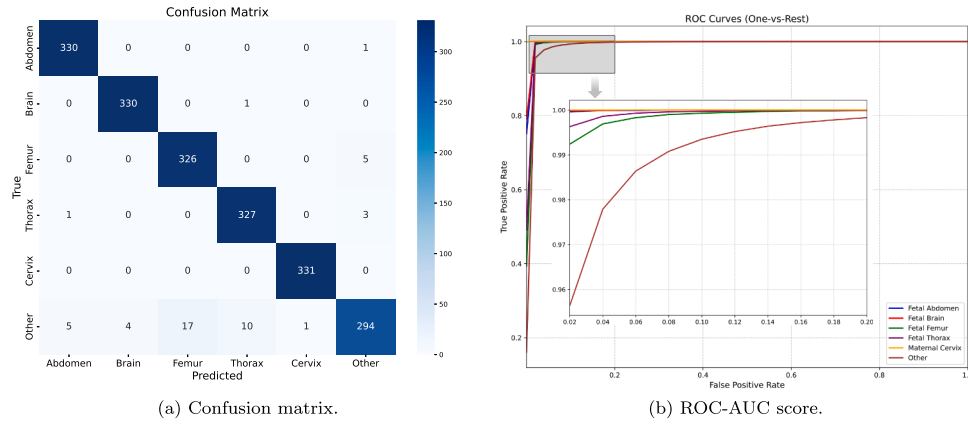


Fig. 7. Comprehensive evaluation results of the proposed model on the test set. (a) Normalized confusion matrix showing the classification performance across the six fetal standard plane classes. The strong diagonal dominance indicates high per-class accuracy, with minor confusion observed primarily involving the 'Other' category. (b) Receiver Operating Characteristic (ROC) curves with One-vs-Rest (OvR) strategy. The Area Under the Curve (AUC) values, all exceeding 0.99, demonstrate the model's excellent binary discriminative power for each class against the rest.

Table 3

Performance comparison of the proposed method with the state-of-the-art models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Specificity (%)
Optimized Feature Selection (Rathika et al., 2024)	97.84	98.03	97.52	97.82	98.16
Ensemble of Deep Learning (Krishna & Kokil, 2024c)	95.69	94.02	96.28	95.08	99.12
Adaptive Channel Weight (Krishna & Kokil, 2024b)	95.33	93.58	95.84	94.64	99.04
Spatial Attention with VGG-19 (Krishna & Kokil, 2024a)	95.40	94.37	95.37	94.85	99.17
Decision Fusion-Based CNN (Sridar et al., 2019)	96.92	96.00	96.98	97.79	98.67
SonoNet-32 (Baumgartner et al., 2017)	93.50	93.29	91.91	92.42	98.54
Ours	98.15	98.12	97.72	98.27	99.59

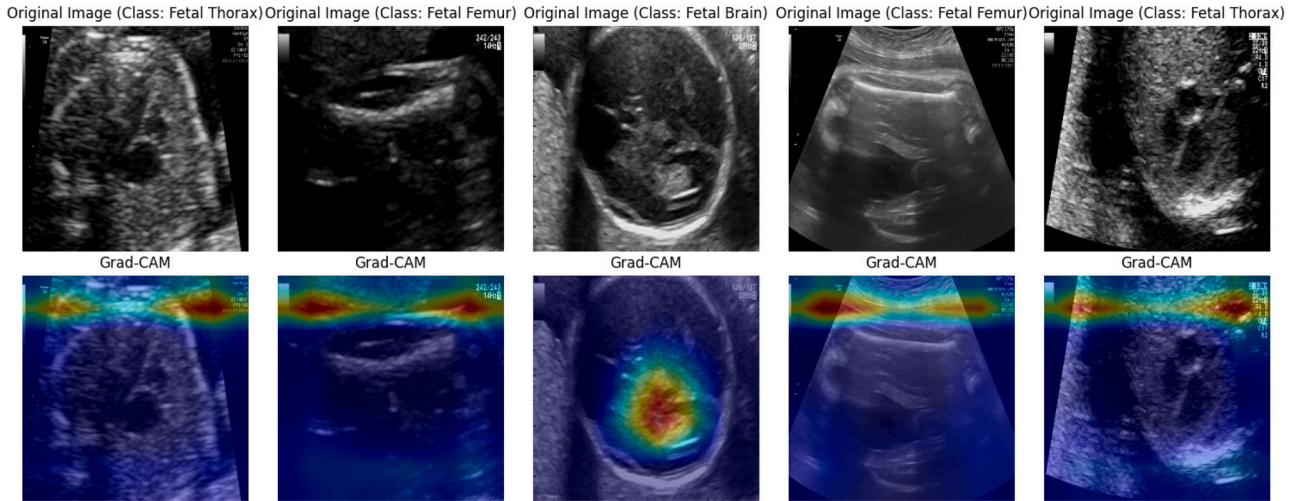


Fig. 8. Visual explanations of model predictions using Gradient-weighted Class Activation Mapping (Grad-CAM). Each row corresponds to a sample from a different fetal standard plane class: (a) Original input ultrasound image; (b) Overlaid Grad-CAM heatmap, where red regions indicate high-activation areas that the model deemed most relevant for its classification decision. The heatmaps show that the model correctly focuses on clinically meaningful anatomical structures, such as the cranial outline in the fetal brain, the curved shape of the femur, and the cardiac region in the thorax, validating its clinical alignment.

highlight its accuracy, reliability, and balanced performance, establishing it as a leading solution for fetal ultrasound plane classification.

4.4. Parameter analysis

To further investigate the influence of the parameter K in the Avg-TopK pooling layer, we conducted a sensitivity analysis to evaluate its effect on model performance using the validation subset. The parameter K determines the number of top activations averaged within each pooling region, thereby controlling the balance between the noise robustness of average pooling and the discriminative selectivity of max

pooling. Notably, when $K = 1$, the operation degenerates to *max pooling*, as only the single highest activation within each pooling window is retained. In contrast, larger K values incorporate multiple high activations, gradually approximating the behavior of average pooling.

We evaluated the enhanced Xception model with K values ranging from 1 to 5 on the validation set, and the results are illustrated in Fig. 10. The performance curves for all evaluation metrics exhibit a consistent trend: model accuracy, precision, and F1-score increase from $K = 1$ to $K = 3$, followed by a gradual decline as K continues to grow. This indicates that a moderate K effectively balances feature selectivity and robustness to noise. The best validation performance 98.15% accuracy,

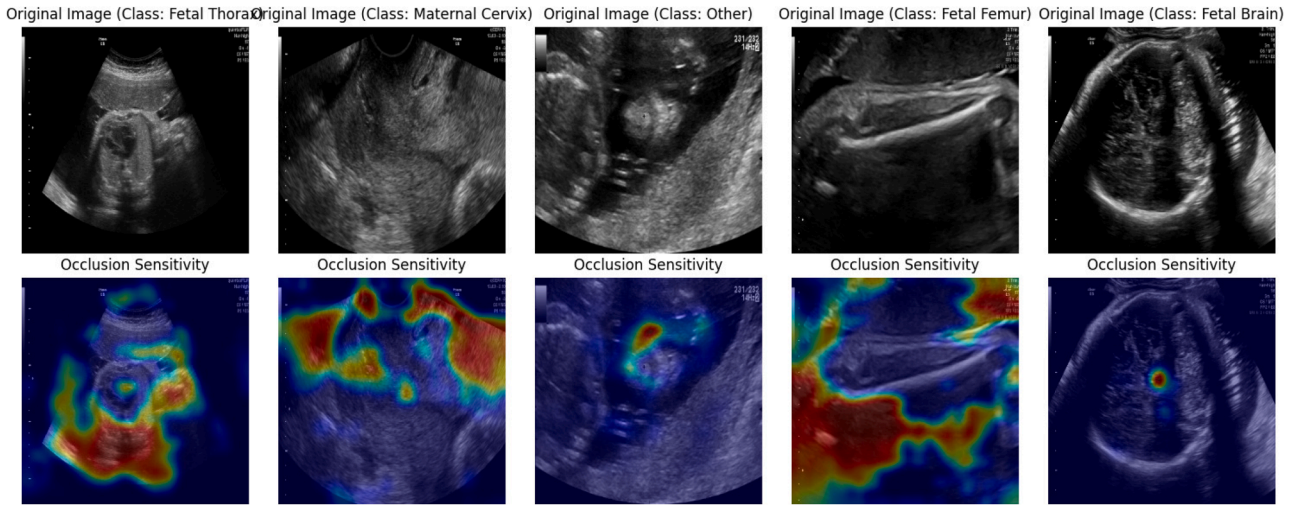


Fig. 9. Interpretability analysis using Occlusion Sensitivity. Each row shows: (a) The original ultrasound image; (b) The corresponding occlusion sensitivity map. A sliding window systematically occludes parts of the image, and the drop in the model's predicted confidence for the true class is recorded. Darker regions in the map indicate areas where occlusion caused a significant confidence drop, meaning these regions are critical for the model's decision. The maps corroborate the Grad-CAM results, showing high sensitivity over key anatomical landmarks and confirming that the model's decisions are based on structurally relevant image content rather than spurious background noise.

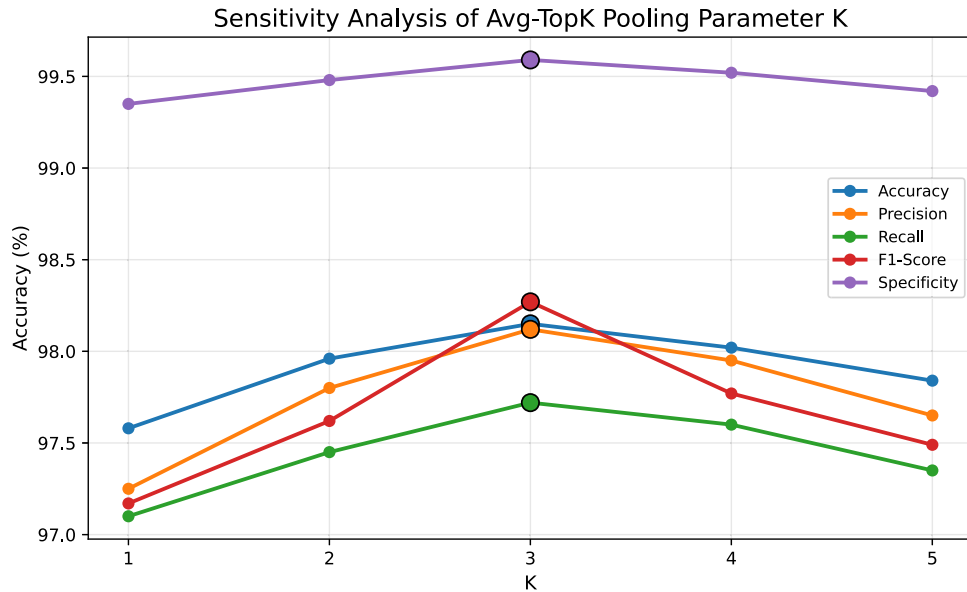


Fig. 10. Sensitivity analysis of the Avg-TopK pooling parameter K conducted on the validation set.

98.12% precision, 97.72% recall, 98.27% F1-score, and 99.59% specificity was achieved at $K = 3$, where the network retains multiple salient activations without diluting discriminative signals. Consequently, $K = 3$ was fixed based solely on validation performance and subsequently applied unchanged during final evaluation on the independent test set. These findings suggest that Avg-TopK pooling with $K = 3$ enables richer representation learning than both max pooling ($K = 1$) and high- K averaging, capturing complementary spatial responses while suppressing background noise.

4.5. Ablation study

To address the scarcity of annotated data, transfer learning was employed, and all pre-trained CNNs were fine-tuned on the target dataset, demonstrating improved generalization and classification performance. Table 4 provides a thorough comparison of various classifiers combined with state-of-the-art pre-trained CNN architectures for standard

plane classification from fetal ultrasound images. The evaluated classifiers include SVM, Decision Tree (DT), k-Nearest Neighbors (KNN), Naive Bayes (NB), and Random Forest (RF). The CNN backbones include AlexNet, ResNet-18, ResNet-50, ResNet-101, VGG-16, VGG-19, MobileNet-v2, ShuffleNet, and Xception. Performance was assessed using accuracy, precision, recall, F1-score, and specificity.

Among the pre-trained backbones, Xception and ResNet-101 demonstrated superior performance across most classifiers, particularly when combined with SVM and RF. For example, ResNet-101 with RF achieved an accuracy of 93.14%, and Xception with SVM achieved 93.50%. These results confirm the strong representation capabilities of deeper architectures and their ability to capture fine-grained anatomical patterns in ultrasound images. However, despite these competitive baselines, the proposed method consistently achieved the best performance across all metrics and classifiers. Specifically, the proposed model achieved an overall accuracy of 98.15%, significantly surpassing the best SOTA combinations. In terms of precision and recall, the model achieved 98.12% and

Table 4

Performance comparison of classifiers combined with state-of-the-art pre-trained CNN architectures and the enhanced Xception.

Metric	Classifier	AlexNet	ResNet-18	ResNet-50	ResNet-101	VGG-16	VGG-19	MobileNet-v2	ShuffleNetV2	Xception	Ours
Accuracy	SVM	93.57	93.17	93.70	93.14	94.55	94.71	93.47	93.31	93.50	98.15
	DT	88.86	87.13	88.54	88.60	92.59	92.61	89.80	88.41	91.33	93.15
	KNN	93.76	92.85	92.98	92.82	94.25	94.29	92.85	93.17	93.64	95.67
	NB	90.30	92.36	92.29	91.41	93.74	91.80	91.46	90.93	91.07	92.37
	RF	93.21	92.75	92.36	93.14	94.19	94.66	92.62	91.90	93.44	97.67
Precision	SVM	92.25	91.49	92.36	92.34	93.09	93.37	92.05	92.46	91.87	98.12
	DT	86.62	83.94	86.94	88.04	88.92	90.92	82.53	80.25	89.61	92.39
	KNN	91.22	90.58	90.95	90.40	91.92	91.90	89.06	91.28	91.36	94.17
	NB	86.58	89.56	89.18	89.00	91.71	89.45	90.15	88.55	89.12	93.49
	RF	91.80	91.38	90.64	91.67	93.33	93.44	91.67	91.56	91.50	95.49
Recall	SVM	92.66	91.02	92.75	91.22	94.56	94.81	92.28	92.66	91.85	97.72
	DT	86.10	83.65	84.59	86.56	90.42	91.26	82.76	79.67	92.33	91.78
	KNN	91.34	90.34	92.67	91.41	92.34	92.33	91.00	92.27	92.54	92.39
	NB	89.80	91.13	90.53	91.41	92.22	91.01	91.22	89.45	89.50	91.79
	RF	91.80	91.03	91.33	93.14	93.56	93.78	90.63	91.76	91.50	97.10
F1-score	SVM	92.45	91.90	92.50	91.75	93.80	93.58	92.16	92.27	91.86	98.27
	DT	86.35	83.80	85.72	87.27	89.64	91.08	82.64	79.94	90.94	90.07
	KNN	92.56	91.73	91.81	91.44	92.13	92.11	90.01	92.27	91.94	93.53
	NB	88.83	90.31	89.85	90.22	91.96	90.20	90.68	88.95	89.31	92.07
	RF	91.89	91.38	90.64	91.62	93.43	93.58	90.94	91.65	91.50	96.10
Specificity	SVM	98.58	92.36	98.64	98.48	98.62	98.46	98.52	96.90	98.60	99.59
	DT	97.57	97.48	98.32	97.47	98.41	97.88	98.10	97.38	97.94	98.10
	KNN	98.09	98.40	97.65	98.33	98.46	98.24	98.12	98.31	98.24	98.64
	NB	98.44	98.47	98.32	98.36	98.26	98.00	98.05	97.56	98.10	96.67
	RF	98.51	98.33	98.64	98.72	98.36	98.56	98.38	98.60	98.47	98.80

97.72%, respectively, indicating excellent reliability in both correctly identifying standard planes and minimizing false positives. The F1-score reached 98.27%, reflecting a balanced performance between precision and recall. Furthermore, the specificity of 99.59% highlights the model's robustness in distinguishing standard planes from non-standard views.

In summary, the experimental results demonstrate that the proposed enhancements to the Xception architecture, including the use of dilated convolutions to expand the receptive field, the integration of channel and spatial attention mechanisms to emphasize diagnostically relevant features, and the Avg-TopK pooling strategy to preserve high-importance activations, significantly improve classification accuracy and robustness. These architectural modifications enable more effective extraction and utilization of semantic information from ultrasound images, resulting in superior performance compared to standard pre-trained models.

The ablation study presented in Fig. 11 evaluates the impact of integrating BSOFS-NBC-based feature selection into the enhanced Xception model across five classifiers, i.e., SVM, DT, KNN, NB, and RF. Performance is assessed using Accuracy, Precision, Recall, F1-score, and Specificity. In addition to architectural improvements, the integration of BSOFS-NBC optimization for feature selection significantly enhanced classification performance by isolating the most relevant features, thereby reducing noise and dimensionality prior to classification. The results clearly demonstrate that incorporating BSOFS-NBC leads to consistent improvements for most classifiers. For example, SVM shows a substantial increase in Accuracy from 94.75% to 98.15%, along with significant gains in Precision (from 92.28% to 98.12%), Recall (from 92.58% to 97.72%), and F1-score (from 92.61% to 98.27%). Similarly, RF achieves an Accuracy improvement from 95.21% to 97.67% and a Precision boost from 92.27% to 95.49%. KNN also benefits from BSOFS-NBC, with Accuracy increasing from 94.32% to 95.67% and F1-score from 92.39% to 93.53%. Although the gains for NB and DT are less pronounced, modest improvements are still observed in several metrics. Notably, Specificity remains high across all configurations, further indicating the robustness of the method in minimizing false positives.

Overall, these results highlight the effectiveness of BSOFS-NBC in enhancing feature selection, leading to better discriminative performance

and generalization capability in fetal ultrasound plane classification. The SVM and Random Forest classifiers, in particular, emerge as the most effective choices when combined with the optimized feature set.

Finally, to quantitatively evaluate the contribution of each component in our enhanced Xception architecture, we conducted a systematic ablation study. We started from a baseline Xception entry flow and incrementally added our proposed modifications. The results, presented in Table 5, clearly demonstrate the performance gain achieved by each component.

The ablation study results, reported as mean $\pm$ standard deviation over 10 independent runs, demonstrate that each component contributes to consistent performance improvements while maintaining low variability across executions. The incorporation of dilated convolutions enhances contextual feature extraction by enlarging the receptive field, resulting in a 1.70% increase in accuracy compared to the baseline, along with stable performance across repeated trials. The addition of attention mechanisms yields the most substantial intermediate improvement, increasing accuracy to 95.80% and specificity to 99.10%, thereby confirming their effectiveness in emphasizing diagnostically relevant regions and suppressing background artifacts.

The integration of the Avg-TopK pooling layer further improves feature robustness by preserving multiple high-activation responses, leading to incremental gains in accuracy and specificity with reduced standard deviation, which reflects enhanced training stability. The final inclusion of the BSOFS-NBC feature selection module achieves the highest overall performance, with 98.15% accuracy and 99.59% specificity. The Wilcoxon signed-rank test ($p = 0.012$), conducted between the Avg-TopK and BSOFS-NBC configurations, indicates that the observed improvement is statistically significant rather than attributable to stochastic variation. The consistent and progressive increase in specificity across model configurations underscores the framework's strengthened capability to correctly identify negative cases.

Table 6 presents an ablation study comparing the proposed Brain Storm Optimization-based Feature Selection with Nearest Better Clustering (BSOFS-NBC) strategy against several widely used and compu-

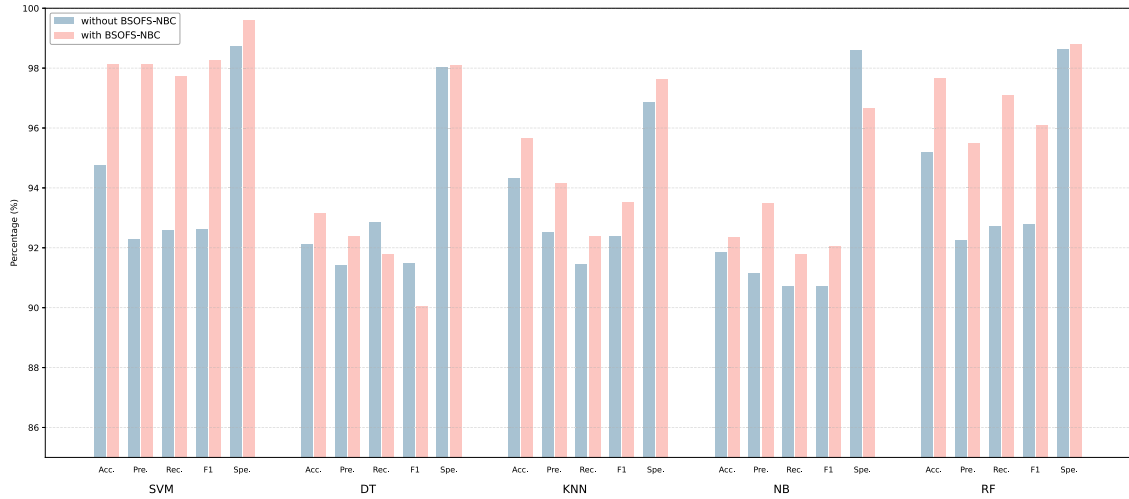


Fig. 11. Ablation study of the feature selection BSOFS-NBC.

Table 5

Component-wise ablation study on the validation set (mean $\pm$ standard deviation over 10 independent runs). Statistical significance is evaluated against the “+ Avg-TopK” configuration using the Wilcoxon signed-rank test.

Model Configuration	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	Spec. (%)
Baseline Xception (Entry Flow Only)	91.50 $\pm$ 0.48	89.25 $\pm$ 0.62	89.80 $\pm$ 0.71	89.52 $\pm$ 0.66	98.25 $\pm$ 0.19
+ Dilated Conv.	93.20 $\pm$ 0.41	91.45 $\pm$ 0.57	91.90 $\pm$ 0.63	91.67 $\pm$ 0.59	98.60 $\pm$ 0.15
+ Attention	95.80 $\pm$ 0.36	94.60 $\pm$ 0.49	94.25 $\pm$ 0.52	94.42 $\pm$ 0.47	99.10 $\pm$ 0.12
+ Avg-TopK	96.45 $\pm$ 0.34	95.80 $\pm$ 0.45	95.50 $\pm$ 0.51	95.65 $\pm$ 0.48	99.35 $\pm$ 0.10
+ BSOFS-NBC (Proposed)*	98.15 $\pm$ 0.29	98.12 $\pm$ 0.38	97.72 $\pm$ 0.44	98.27 $\pm$ 0.41	99.59 $\pm$ 0.08

*Statistically significant improvement over “+ Avg-TopK” (Wilcoxon signed-rank test, $p = 0.012$).

Table 6

Ablation study comparing BSOFS-NBC with standard feature selection methods.

Feature Selection Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Specificity (%)
No Feature Selection	94.45	96.32	95.88	96.10	98.94
PCA	95.98	96.85	96.40	96.62	99.02
LASSO	96.21	97.05	96.74	96.89	99.10
RFE	96.64	97.52	97.18	97.33	99.28
BSOFS-NBC (Proposed)	98.15	98.12	97.72	98.27	99.59

tationally efficient feature selection methods, including Principal Component Analysis (PCA), Least Absolute Shrinkage and Selection Operator (LASSO), and Recursive Feature Elimination (RFE), as well as a baseline without feature selection. The results demonstrate that incorporating conventional feature selection techniques yields noticeable improvements over the no-selection baseline, confirming the importance of reducing redundant and less informative features. Principal Component Analysis (PCA), a linear projection-based dimensionality reduction method, and Least Absolute Shrinkage and Selection Operator (LASSO), a sparsity-inducing regularization technique, provide moderate performance gains, while Recursive Feature Elimination (RFE), which follows a greedy backward selection strategy, further enhances classification performance. However, the proposed BSOFS-NBC consistently achieves superior results across all evaluation metrics, attaining the highest accuracy (98.15%), precision (98.12%), recall (97.72%), F1-score (98.27%), and specificity (99.59%). Unlike the conventional approaches that rely on linear assumptions or sequential elimination, BSOFS-NBC leverages a global search mechanism to identify more discriminative and complementary feature subsets. Although the optimization process introduces additional computational cost during training, it is performed offline and does not impact inference efficiency. These findings substantiate the necessity of the proposed optimization strategy in achieving improved

classification performance while maintaining practical deployment feasibility.

4.6. Computational complexity analysis

Beyond accuracy, computational efficiency is a crucial consideration for potential deployment in clinical settings. Table 7 compares the complexity of our proposed framework against several baseline models, such as: VGG-19, ResNet-101, Xception, MobileNet-v2, MobileNet-v3, EfficientNet-B0, and ShuffleNetV2, including the number of parameters, floating-point operations (FLOPs) for a single forward pass, and average inference time per image measured on our experimental setup.

The results demonstrate that conventional deep models such as VGG-19 (143.7M parameters) and ResNet-101 (44.6M parameters) incur substantially higher computational costs without achieving superior classification performance. While lightweight architectures like MobileNet-v3 (95.10%) and EfficientNet-B0 (95.80%) provide improved parameter efficiency (5M parameters), their accuracy remains lower than that of the proposed approach. ShuffleNetV2 achieves the lowest computational burden (2.3M parameters, 0.15 GFLOPs), but at the expense of reduced accuracy (93.31%). In comparison, the proposed Enhanced Xception Entry Flow achieves 96.45% accuracy with only 4.8M parameters

Table 7
Computational complexity comparison of different models.

Model	Accuracy (%)	Parameters (M)	FLOPs (G)	Inference Time (ms)
VGG-19	94.71	143.7	19.6	45.2
ResNet-101	93.14	44.6	7.8	28.5
MobileNet-v2	93.47	3.5	0.3	12.1
MobileNet-v3	95.10	5.4	0.2	11.4
EfficientNet-B0	95.80	5.3	0.4	13.2
ShuffleNetV2	93.31	2.3	0.15	9.8
Xception (Full)	93.64	22.9	5.3	20.8
Baseline Xception (Entry Flow Only)	91.50	4.2	1.8	8.5
Proposed (Enhanced Xception Entry Flow)	96.45	4.8	2.1	9.2
Proposed (Enhanced Xception + BSOFs-NBC)	98.15	4.8	2.1	12.5

and 2.1 GFLOPs, demonstrating a favorable balance between predictive performance and model complexity. Furthermore, the integration of BSOFs-NBC further improves accuracy to 98.15% while maintaining the same parameter size, with only a modest increase in inference time. These results confirm that the proposed framework provides a superior accuracy-efficiency trade-off compared to both conventional deep networks and established lightweight architectures.

5. Discussions

This study proposes a hybrid framework that integrates an enhanced Xception network with BSOFs-NBC for automatic classification of fetal ultrasound standard planes. The model achieves an overall accuracy of 98.15%, surpassing previously reported approaches. While these results are promising, a deeper interpretation is necessary to elucidate the sources of improvement, identify remaining weaknesses, and discuss the broader clinical significance.

The superior performance can be primarily attributed to the synergistic interaction between the two stages of the framework. The enhanced Xception network benefits from dilated convolutions that enlarge the receptive field, enabling the model to capture contextual cues from broader anatomical regions—an essential factor in distinguishing visually similar planes such as the abdomen and thorax. Furthermore, the embedded attention mechanism effectively suppresses speckle noise and irrelevant backgrounds, emphasizing diagnostically meaningful structures. These effects are corroborated by the Grad-CAM visualizations (Fig. 8), which show activation patterns concentrated over clinically relevant anatomical landmarks, such as the femoral shaft and cranial vault. The subsequent feature optimization by BSOFs-NBC further contributes by removing redundant features and promoting generalizable representations, as supported by the ablation results (Fig. 11). Together, these components suggest that the framework not only enhances discriminative capability but also reduces overfitting and computational complexity.

Nevertheless, the analysis of error cases reveals important weaknesses. The confusion matrix (Fig. 7a) indicates that the majority of misclassifications occur in the heterogeneous “Other” category, frequently being confused with “Fetal Femur” or “Fetal Thorax.” A plausible explanation is that non-standard or transitional views often contain partial anatomical structures, leading to ambiguous visual patterns that challenge even experienced sonographers. This suggests that a rejection or uncertainty-aware mechanism, rather than forced single-class assignment, may be more suitable for handling ambiguous frames in real-time screening scenarios. More generally, this observation implies that a purely frame-based classification approach may be insufficient for handling the continuous nature of real-world ultrasound video, where temporal and contextual information is critical for accurate interpretation.

From a clinical perspective, the model’s high specificity (99.59%) is encouraging, as it reduces false-positive detections and may help avoid unnecessary follow-up examinations. The strong recall for key standard planes suggests potential utility in assisting practitioners during biomet-

ric assessment. However, given the observed limitations in transitional frames and the absence of temporal modeling, the current system should be regarded as a decision-support tool rather than a fully autonomous screening solution. Further validation in real-time clinical workflows is required before practical deployment.

Although transformer-based and hybrid CNN-Transformer architectures have recently demonstrated strong capability in modeling long-range dependencies and global spatial-semantic relationships, their effectiveness typically relies on large-scale annotated datasets and substantial computational resources. In fetal ultrasound imaging, however, data are often limited, imbalanced, and affected by speckle noise and anatomical ambiguity, conditions under which data-intensive global attention mechanisms may suffer from unstable optimization and reduced generalization. In contrast, Xception benefits from strong convolutional inductive biases and depthwise separable convolutions, enabling efficient and stable learning of fine-grained local anatomical structures while maintaining low parameter complexity and fast inference. By coupling Xception with the BSOFs-NBC feature selection strategy, the proposed framework further enhances discriminative capability by explicitly removing redundant features and promoting generalizable representations, thereby achieving competitive or superior performance compared to global attention models at a fraction of the computational cost. Importantly, the technical contribution of this work lies in establishing a discriminative representation-selection framework for ultrasound plane classification under data-constrained conditions. The proposed system is designed as a robust frame-level foundation upon which temporal modeling and uncertainty-aware modules can be integrated in future work.

Despite these encouraging results, several limitations must be acknowledged. First, the model was trained and validated on a curated dataset with high-quality annotations but noticeable class imbalance, particularly in the “Other” category, which does not fully represent the variability encountered in routine clinical practice. Consequently, its performance in uncurated, real-time ultrasound video streams remains uncertain. Second, although the model demonstrates strong frame-level performance, reliable clinical deployment in continuous ultrasound workflows requires integration of temporal consistency modeling and uncertainty-aware mechanisms, particularly for handling transitional or ambiguous frames categorized as “Other.” Third, the current framework is limited to static image classification; to provide direct diagnostic utility, it must evolve toward real-time video analysis and anatomical localization. Finally, the model’s generalizability across different ultrasound devices, patient populations, and imaging settings remains to be established. Addressing these limitations will require large-scale, multi-center validation studies and more robust strategies for handling domain shifts.

6. Conclusion

This study effectively addressed the challenge of fetal ultrasound standard plane classification through a novel dual-stage framework. The key contributions include an enhanced Xception architecture for discriminative feature extraction and a BSOFs-NBC algorithm for optimal feature selection. Experimental evidence demonstrates the frame-

work's superiority, achieving state-of-the-art performance with 98.15% accuracy and outperforming established CNN and ensemble baselines across all evaluation metrics. The ablation study further verifies that each architectural component contributes significantly to overall performance. Overall, this work provides methodological innovation and demonstrates promising potential for supporting intelligent decision-assistance systems in prenatal imaging.

While the proposed framework establishes an efficient frame-level foundation, several directions remain critical for future research. Nevertheless, its current design focuses on static image classification, and additional validation with temporal modeling is required before full integration into real-time clinical screening workflows. These include extending the model to continuous ultrasound video analysis by incorporating temporal modeling and uncertainty-aware decision mechanisms, integrating anatomical localization or object detection for direct biometric measurement, and validating the approach on multi-center, multi-vendor datasets to improve robustness and clinical generalizability. In addition, domain generalization and harmonization strategies, such as intensity normalization, adversarial domain adaptation, and federated learning across institutions, represent promising avenues for reducing scanner-specific bias and facilitating broader clinical deployment. Moreover, we will also integrate large-scale and multimodal evolutionary optimization paradigms (Xu et al., 2023; Zhu et al., 2026) into feature selection or hyperparameter tuning to enhance robustness, mitigate overfitting, and improve scalability to complex, heterogeneous, and multi-center clinical data.

CRedit authorship contribution statement

Javed Hossain: Conceptualization, Methodology, Formal analysis, Writing – original draft, Visualization; **Peilan Xu:** Conceptualization, Supervision, Project administration, Writing – review & editing, Funding acquisition; **Yong Liu:** Software, Data curation, Validation, Investigation, Visualization; **Jingwen Li:** Resources, Writing – review & editing; **Ziqian Kong:** Investigation, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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